

5. Radioactive isotopes have a wide variety of uses.

(a) (i) State what is meant by the term ‘isotopes’. [2]

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(ii) State why some isotopes are radioactive. [1]

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(b) One use of radiation is in a smoke detector.

The radiation ionises the air inside the detector and causes a current that is detected. If smoke particles are present they block the radiation so that a current is no longer detected. This sets off the alarm.

The table gives information about different radioisotopes.

isotope	half-life	type of emitter
americium-241	432 years	alpha
radium-224	3.6 days	alpha
caesium-137	30 years	beta and gamma

Explain why americium-241 is suitable for this use but radium-224 and caesium-137 are not. [3]

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- (c) Brachytherapy is a form of radiotherapy used in treating prostate cancer. Pellets containing radioactive material are planted inside the prostate. One patient has 75 pellets containing iodine-125 planted in his prostate. Each pellet has an initial activity of 16 MBq.

- (i) The iodine-125 has a half-life of 60 days.
Calculate the time taken for the total activity of the iodine-125 in the 75 pellets to drop to 37.5 MBq. [3]

time = days

- (ii) Iodine-125 is a gamma emitter.
For 60 days after his treatment the patient is warned not to stay too close to people for long periods of time.
Explain why the gamma radiation may pose a risk to other people. [2]

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